# Correlated Randomly Growing Graphs


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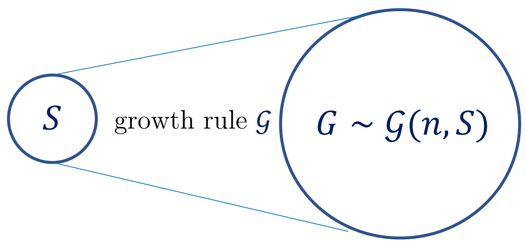
## A new model of correlated growing graphs

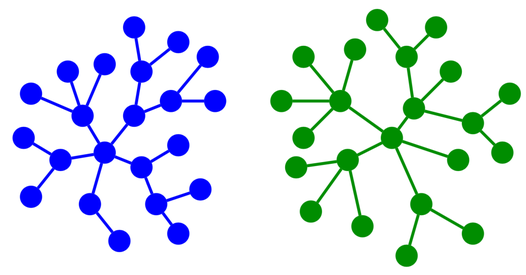
**Randomly growing graphs**

- Start from seed graph $S$
- At each time step add a new vertex along with new edges

Canonical examples:

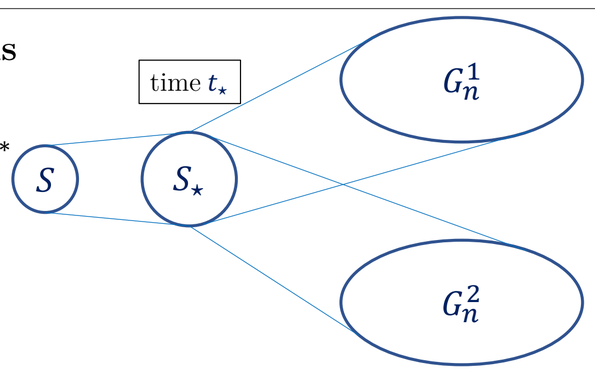
- uniform attachment (UA)
- preferential attachment (PA)

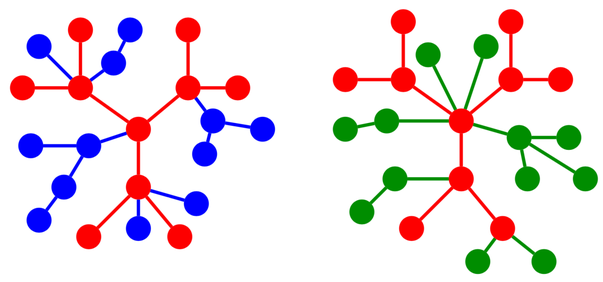


**Correlated growing graphs**

- Start from seed graph $S$
- Grow together until time $t_*$
  - $S_* \sim \mathcal{G}(t_*, S)$
- Then grow independently
  - $G_n^1 \sim \mathcal{G}(n, S_*)$
  - $G_n^1 \sim \mathcal{G}(n, S_*)$
- $(G_n^1, G_n^2) \sim \mathcal{CG}(n, t_\star, S)$



**Fundamental Questions:**

1. **Detection:** Can we distinguish between the hypotheses
$$H_0 : (G_n^1, G_n^2) \sim \mathcal{G}(n,S)^{\otimes 2}$$
$$H_{t_\star} : (G_n^1, G_n^2) \sim \mathcal{CG}(n, t_\star, S)$$
with positive power or high probability?
2. **Estimation:** $(G_n^1, G_n^2) \sim \mathcal{CG}(n, t_\star, S)$ where $t_*$ is unknown. Can one estimate $t_*$ and the shared subgraph $S_*$ even as $n \to \infty$?

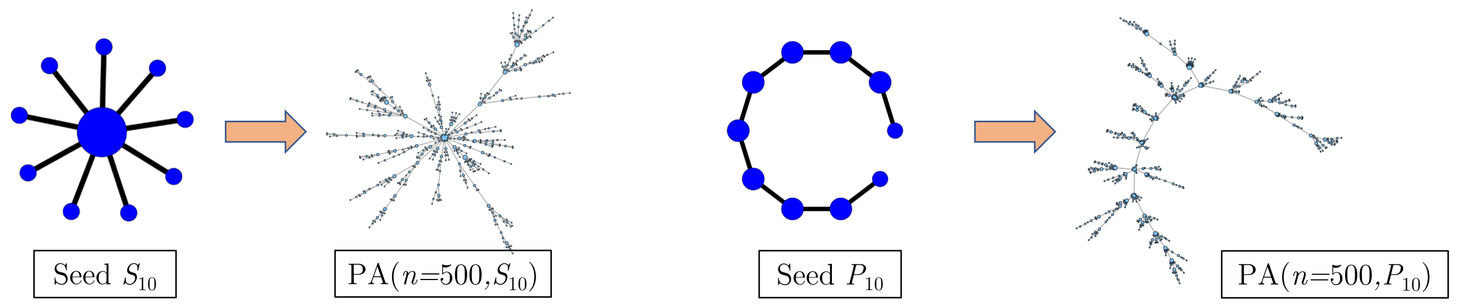
## Main Result 1: Detection with Positive Power

**Theorem (Rácz, Sridhar 2020).** Fix a seed graph $S$, correlation time $t_*$ and a Markov attachment rule $\mathcal{G}$. Suppose there are graphs $G$, $G'$ on $t_*$ vertices such that $\mathbb{P}_S(G), \mathbb{P}_S(G') > 0$ and
$$\lim_{n\to\infty} TV(\mathcal{G}(n,G), \mathcal{G}(n,G')) > 0.$$
Then
$$\lim_{n\to\infty} TV(\mathcal{G}(n,S)^{\otimes 2}, \mathcal{CG}(n,t_\star,S)) > 0.$$

In other words: **if the seed has influence, detection is possible.** In UA and PA trees, the seed always has influence [1-3].

**Corollary.** Let $S$ be a finite tree with at least 2 vertices and let $t_* > max\{|S|, 3\}$. Then
$$\lim_{n\to\infty} TV(PA(n,S)^{\otimes 2}, CPA(n,t_\star,S)) > 0.$$
Furthermore, the same holds for UA.

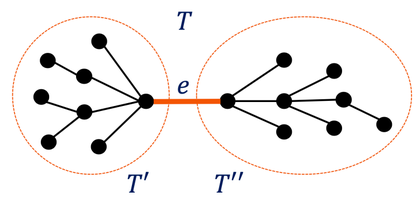
## Statistics for detecting correlation




**Detecting correlation in PA trees.**

$$A_{u,n} := \left\{ \frac{\Delta(T_1(n))}{\sqrt{n}} > u, \frac{\Delta(T_2(n))}{\sqrt{n}} > u \right\}$$
$$\lim_{n\to\infty} \mathbb{P}_0(A_{u,n}) \sim \frac{16}{\pi} u^{-2} \exp(-u^2/2)$$
$$\lim_{n\to\infty} \mathbb{P}_{t_\star}(A_{u,n}) \sim \frac{2^{t_\star+1}}{\pi} u^{-2} \exp(-u^2/2).$$

**Detecting correlation in UA trees.**




$$h(T,e) := \frac{|T'|^2 |T''|^2}{|T|^4} \qquad H(T) := \sum_{e \in E(T)} h(T,e)$$

In PA and UA trees, correlation can be detected with positive power **even if the graphs are grown together for a single time step!**

## Main Result 2: Detection with high probability

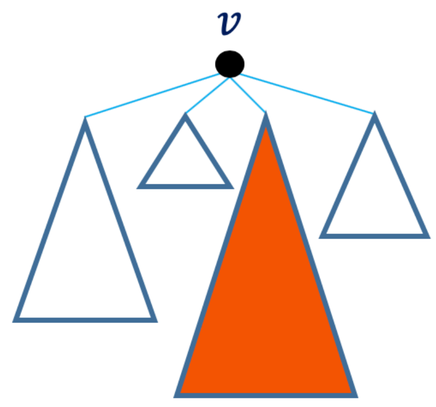
**Theorem (Rácz, Sridhar 2020).** Fix a seed graph $S$ and a Markov attachment rule $\mathcal{G}$. Suppose there is a function $f$ such that $\lim_{n\to\infty} f(G_n) =: f_\infty$ exists a.s. and is an absolutely continuous random variable. Then
$$\lim_{t_\star\to\infty} \lim_{n\to\infty} TV\left(\mathcal{G}(n,S)^{\otimes 2}, \mathcal{CG}(n,t_\star,S)\right) = 1.$$

For UA and PA trees, we can let $f$ be the centrality measure described on the right.

**Corollary.** Let $S$ be a finite tree with at least 2 vertices. Then
$$\lim_{t_\star\to\infty} \lim_{n\to\infty} TV\left(PA(n,S)^{\otimes 2}, CPA(n,t_\star,S)\right) = 1.$$
Furthermore, the same holds for UA.

## Centrality in trees

The *anti-centrality* of a vertex $v$ is the size of its largest subtree:
$$\Psi_T(v) := \max_{u \in \mathcal{N}_v(T)} |(T,v)_{u\downarrow}|$$




The *limiting anti-centrality* of a sequence of PA/UA trees $\{T_n\}_n$ exists a.s. and is an absolutely continuous random variable:
$$\Psi(v) := \lim_{n\to\infty} \frac{1}{n} \Psi_{T_n}(v)$$

The centroid is the vertex with minimum anti-centrality. In PA/UA trees the centroid $\theta_n$ is *persistent* [4] so the following quantity exists a.s. and is absolutely continuous:
$$\lim_{n\to\infty} \frac{1}{n} \Psi_{T_n}(\theta_n) = \lim_{n\to\infty} \frac{1}{n} \min_{v \in T_n} \max_{u \in \mathcal{N}_v(T_n)} |(T_n, v)_{u\downarrow}|$$

The proof idea is to express the centrality measures as combinations of Pólya Urns.

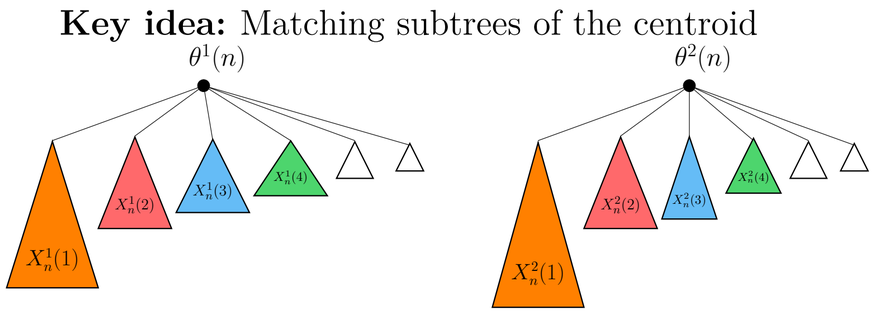
## Main Result 3: Estimation

**Theorem (Rácz, Sridhar 2020).** Consider $(T_n^1, T_n^2) \sim CPA(n, t_*, S)$. There is an estimator
$$\hat{t} = \hat{t}\left(T_n^1, T_n^2\right)$$
computable in poly time, such that
$$\lim_{t_\star\to\infty} \liminf_{n\to\infty} \mathbb{P}\left(\left(1 - \frac{\log\log t_\star}{\sqrt{\log t_\star}}\right) t_\star \le \hat{t}_n \le \left(1 + \frac{\log\log t_\star}{\sqrt{\log t_\star}}\right) t_\star\right) = 1.$$
The same holds for UA trees.

In other words, we can estimate $t_*$ with **vanishing relative error**, even as $n \to \infty$.

## Proof idea

**Key idea:** Matching subtrees of the centroid




- Idea behind estimator: $(X_n^1(k) - X_n^2(k))^2 \asymp \frac{1}{t_\star}$
- **Coarse estimate** obtained by matching largest subtrees of the centroid
$$\frac{t_\star}{\log t_\star} \le \hat{t} \le t_\star \log t_\star$$
- **Finer estimate** obtained by averaging over the largest $\sim \log t_\star$ subtrees

## Acknowledgements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## References


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